

# Hackathon Challenge Theme — AI for Next Billion

(AIMS @ DTU)

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## Problem Statement

**Make AI truly accessible for everyone.**

Build a solution that can realistically reach 1 billion people (or be on a credible path to it) and meaningfully improve daily life—especially for users with limited connectivity, limited English proficiency, limited device storage, limited digital literacy, or accessibility needs.

### What “accessible” means (your north star)

Your product should reduce at least one major barrier:

- **Connectivity:** works on low bandwidth / intermittent internet / offline-first
  - **Cost:** low compute, low token usage, low per-user cost
  - **Device constraints:** runs on mid/low-end phones
  - **Language:** supports local languages (speech + text)
  - **Literacy:** voice-first, guided flows, minimal typing
  - **Disability access:** screen-reader friendly, captions, color contrast, large text
  - **Trust:** clear sources, safe defaults, privacy-aware
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## Required: Make it Specific (No Stale Submissions)

This theme is popular. To avoid generic projects, your submission must clearly specify:

1. **Target users** (who exactly?)
  2. **One concrete job-to-be-done** (what task improves?)
  3. **Distribution channel** (how will it reach people?)
  4. **Cost model** (why can it scale?)
  5. **One measurable metric** (success = what?)
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## Required: Agentic Automation (The Differentiator)

Your solution must include agentic automation—not just chat.

**Minimum bar:**

- The system can take actions using at least one tool/API (e.g., messaging, phone/SMS gateway, calendar, forms, web fetch, database, file generation, government/utility services, etc.)
- Users can trigger a repeatable workflow, not only ask questions

#### Examples:

- “Send an SMS reminder, schedule a follow-up, and fill a form”
- “Translate + summarize a document, then generate a reply and send it”
- “Collect info from the user over voice, then submit an application draft”

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## Guidance: Do’s (What We Want)

Pick a real-world surface area where scale is plausible:

- **Offline-first assistants** (downloaded model / on-device inference / cached knowledge)
- **SMS/WhatsApp/IVR AI** (works on feature phones or low-end phones)
- **Local-language copilots** (speech-to-speech in Hindi + regional languages)
- **Community worker tools** (ASHA/Anganwadi/teachers/field agents)
- **Government/utility form helpers** (actionable workflows, not just advice)
- **Accessibility tools** (vision/hearing/motor impairments)

#### Strong signals to include:

- Graceful degradation when internet is slow
- Clear UX for low literacy users
- Privacy-by-default (don’t collect what you don’t need)

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## Guidance: Don’ts (To Keep the Bar High)

Submissions will be disqualified or heavily penalized if they are primarily:

1. A generic chatbot about any topic
2. Prompt-only wrappers (something you can do by prompting a public model directly)
3. Basic RAG / generic context management systems without a strong product layer
4. Medical advice products (diagnosis/treatment recommendations)
  - *Note: healthcare-adjacent workflows are allowed only if they are non-medical and operational (e.g., appointment scheduling, information routing, reminders, translation), with appropriate disclaimers.*

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## Suggested Directions (Concrete, Not Generic)

Use these as inspiration—teams should still narrow to a specific user + job.

1. **Offline/Low-bandwidth AI companion for farmers/SMEs**
  - Inputs: voice notes + photos (when available)
  - Outputs: step-by-step actions, local language, SMS follow-ups
  - Actions: reminders, market price fetch, logistics coordination
2. **SMS-based “forms agent”**
  - Guides users through government/utility forms over SMS/WhatsApp
  - Generates a completed draft / checklist
  - Actions: create a PDF, send to user, schedule follow-up
3. **Local-language job search + application agent**
  - Voice-first resume builder
  - Auto-applies to listings / drafts messages
  - Actions: email/WhatsApp messages, calendar scheduling
4. **Accessibility-first reading + doing**
  - Reads documents aloud, translates, extracts tasks
  - Actions: create reminders, send summaries, fill forms

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## Deliverables

Your team should submit:

- **Demo** (live or local)
- **Pitch** (2–3 min)
- **README** describing:
  - target user + job
  - distribution channel
  - cost/scaling logic
  - what tools/actions the agent performs
  - what you explicitly did NOT build

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## Judging Rubric (100 Points)

| Points | Category                           | Criteria                                                                                                    |
|--------|------------------------------------|-------------------------------------------------------------------------------------------------------------|
| 35     | <b>Impact + Reach Plausibility</b> | Clear target user; credible path to large-scale adoption; meaningful improvement vs status quo.             |
| 25     | <b>Product Specificity + UX</b>    | Not generic; works under constraints (low bandwidth, low literacy); simple, trustworthy interaction design. |

|           |                                   |                                                                                          |
|-----------|-----------------------------------|------------------------------------------------------------------------------------------|
| <b>20</b> | <b>Agentic Automation Quality</b> | Takes real actions (tools); workflows are repeatable; failure handling and user control. |
| <b>10</b> | <b>Technical Execution</b>        | Robustness, latency, offline strategy; sensible architecture.                            |
| <b>10</b> | <b>Originality</b>                | Fresh angle, not a clone.                                                                |

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## Quick Checklist (Before You Present)

- Can a non-technical user understand it in 30 seconds?
- Does it still work (even partially) with bad internet?
- Does it do something, not just talk?
- Is it safe and non-medical?
- Do you have one metric you can claim improvement on?